Knowledge Distillation for Building Lightweight Deep Learning Models in Visual Classification Tasks

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Introduction

Knowledge Distillation (KD): It is the process of distilling or transferring the knowledge from a (set of) large, cumbersome model(s) to a lighter, easier-to-deploy single model, without significant loss in performance [1].

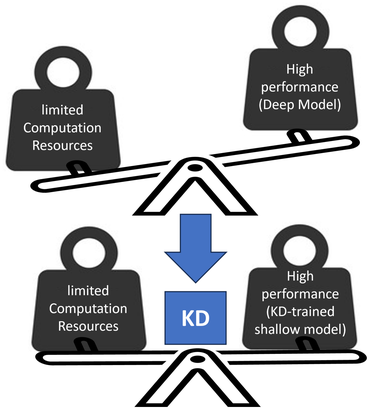


Fig. 1 Challenging problem addressed by KD

Motivation

Goals:

- **Compression**: To reduce the size of a complex, deep model without substantially sacrificing accuracy, making the distilled model more storage-efficient.
- **Speed**: To enhance the inference speed of the model, allowing for faster predictions which is crucial for time-sensitive applications.
- **Energy Efficiency**: To lower the energy requirements for running the model, thus saving battery life on mobile devices and reducing operational costs on servers.
- **Efficient deployment**: To facilitate deployment of the model on devices with limited computational resources, such as smartphones and IoT devices.

Previous methods for model compression includes [2]:

- **Pruning**: Eliminates less important weights or neurons to simplify the model.
- **Quantization**: Reduces the numerical precision of weights and activations, (e.g., 32-bit to 8-bit), to shrink model size.
- **Weight Sharing**: Clusters weights to share a common value.
- **Sparse Representations**: Promotes sparsity in weights during training, resulting in many weights being near zero, which are more storage-efficient.

Advantages of KD over previous methods:

- **Preserving Performance:** By training the student model to mimic the output distribution of the teacher model, KD can capture the information present in the teacher model.
- **Architectural Flexibility**: KD allows for the transfer of knowledge from any type of large model (teacher) to any type of smaller model (student), irrespective of their internal architectures.

Methods

Databases:

- **MNIST:** Multi-class digital classification of handwritten digits from 0 to 9 (10 classes) – (60k training examples, 10k test examples)
- **MHIST:** Binary-class texture analysis in cancer histology (two classes; benign and pre-cancerous - 2175 training examples, 977 test examples – imbalanced database)



Fig. (2a) MNIST

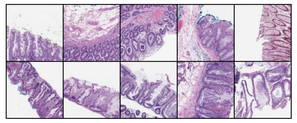


Fig. (2b) MHIST

KD System Overview

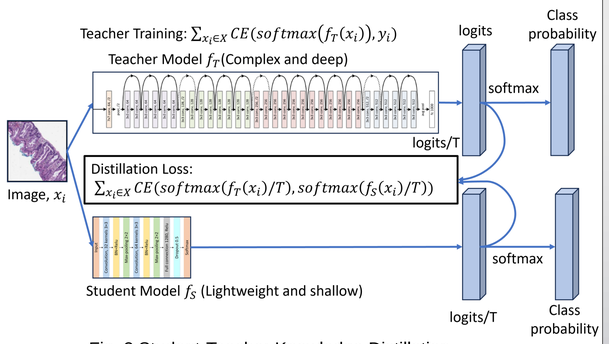


Fig. 3 Student-Teacher Knowledge Distillation

SOTA: Teaching Assistant (TA) [4]

- Bridge the knowledge gap between teacher and student model effectively, especially if the architecture capacity difference is large.
- Gradual reduction capacity gap can lead to possible performance improvement of student model.

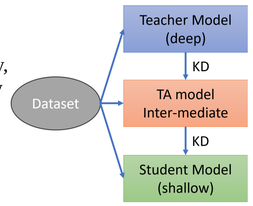


Fig. 4 TA-based KD

Transfer Learning (TL):

TL differs from KD in the following aspects:

- Useful in training deep models on small scale datasets, e.g. MHIST
- Model compression is not the goal here, as the same pretrained model is used except for final classification stage.

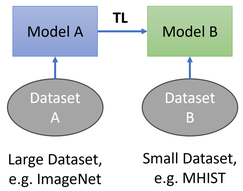


Fig. 5 TL framework

Results

MNIST Results:

Table 1. Comparing Flops and number of Trainable parameters for teacher and student models

	Teacher Model	Student Model
Trainable parameters (M)	1.01	1.24
Flops (M)	11.02	1.24

Table 2. Student test accuracy (%) with and without KD

Teacher	Student (KD) $\alpha = 0.5, T = 64$	Student (no KD)
99.26	**98.9**	98.14

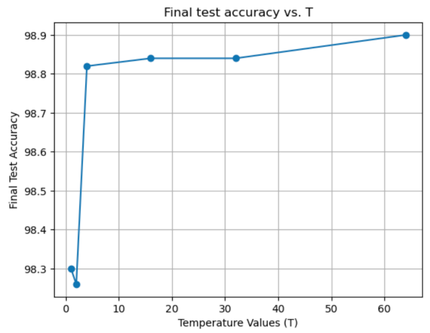


Fig. 6 Student test accuracy vs KD temperature

MHIST Results:

Table 3. Performance of teacher and student models using different training schemes

Models	F1 score	test accuracy (%)
Teacher Model (Scratch)	0.59	73.22
Teacher Model (TL)	0.788	85.32
Student Model (scratch)	0.607	75.8
Student Model (TL)	0.78	84.48
Student Model (TL + KD)	**0.798**	85.46

Table 4. Number of trainable parameters and flops in Million (M) for the teacher, TA, and the student model

	Trainable parameters (M)	Flops (M)
Teacher Model (ResNet50V2)	25.66	3.48k
TA Model (EfficientNetB2)	8.49	670.8
TA Model (EfficientNetB3)	11.57	979.17
TA Model (DenseNet121)	7.56	2.385k
Student Model (MobileNetV2)	2.91	300.18

Table 5. F1 score of the test set using TA-based KD for different TA pre-trained architectures

	TA		Student	
TA Model	F1 score	Accuracy %	F1 score	Accuracy %
EfficientNetB2	0.7375	82.57	0.783	85.16
EfficientNetB3	0.761	83.32	0.799	84.933
DenseNet121	0.772	83.97	0.742	83.6

Conclusions

- Knowledge Distillation is a strategy to transfer knowledge from a complex (teacher) model to a simpler (student) model.
- It preserves performance while enabling deployment on resource-limited computing platforms.
- Training Student models using KD showed comparable performance with their corresponding teacher models.
- In addition, noticeable performance improvement is achievable compared to training student from scratch (no KD).
- Introducing Teacher Assistant (TA) models improves distillation efficacy by bridging knowledge gap between teacher and student model.
- Our results didn't show significant performance improvement using TA which might be attributed to the following reasons:
  - TA model might not introduce any new or beneficial information beyond what the teacher model already provides.
- TA requires careful selection of the model architecture, hence it does not always guarantee performance improvement

Bibliography

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